



RV College of  
Engineering®



**Subject: Computer Networks & Protocols**

**PBL Topic: Adaptive Wi-Fi Bandwidth Allocation  
Using Machine Learning**



RV College of  
Engineering®

*Go, change the world®*

# Team Members

| USN        | NAME          | Email Id                     |
|------------|---------------|------------------------------|
| 1RV24EC404 | Gurukiran B H | gurukiranbh.ec24@rvce.edu.in |
| 1RV23EC090 | Nishaanth K S | nishaanthks.ec23@rvce.edu.in |



RV College of  
Engineering®

# Introduction

*Go, change the world®*

Wireless Local Area Networks (Wi-Fi) have become an essential part of modern communication, connecting a large number of devices such as smartphones, laptops, and IoT devices. As the number of connected users increases, the network often experiences congestion, resulting in higher delay, increased packet loss, and reduced throughput, which affect the Quality of Service (QoS). Traditional bandwidth allocation methods rely on fixed rules and are unable to adapt to changing network conditions. To address this problem, this project proposes a Machine Learning-based adaptive Wi-Fi bandwidth allocation system that predicts network priority using Delay, Packet Loss, and Throughput. The system compares the performance of Decision Tree and Random Forest algorithms, and based on the predicted priority, recommends appropriate bandwidth allocation to improve overall network performance and user experience.



RV College of  
Engineering®

# Problem Statement

*Go, change the world®*

Traditional Wi-Fi bandwidth allocation uses fixed rules and cannot adapt to changing network conditions. This results in increased delay, packet loss, and reduced throughput during network congestion. Therefore, an intelligent system is needed to dynamically predict network priority and allocate bandwidth efficiently using Machine Learning.



# Literature Survey

| Sl.No. | Paper Title, Journal Name,<br>Year                                                                                                                        | Summary                                                                                                             |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 1.     | <b>Federated Reinforcement Learning for Distributed MAC Optimization in IEEE 802.11bn Networks</b> , <i>Discover Electronics (Springer)</i> , <b>2026</b> | Uses Federated Reinforcement Learning to optimize MAC scheduling and improve QoS in next-generation Wi-Fi networks. |
| 2.     | <b>Machine Learning-Driven Analysis of User Bandwidth Allocation and Performance in 5G Network</b> , <i>IEEE Access</i> , <b>2025</b>                     | Applies machine learning to optimize user bandwidth allocation and improve QoS in 5G networks                       |



# Literature Survey

| Sl.No. | Paper Title, Journal Name, Year                                                                                                                                         | Summary                                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 3.     | <b>Hybrid-Task Meta-Learning: A GNN Approach for Scalable and Transferable Bandwidth Allocation</b> , <i>IEEE Transactions on Wireless Communications</i> , <b>2024</b> | Proposes a Graph Neural Network (GNN) with meta-learning for scalable wireless bandwidth allocation under varying QoS requirements. |
| 4.     | <b>Machine Learning &amp; Wi-Fi: Unveiling the Path Towards AI/ML-Native IEEE 802.11 Networks</b> , <i>Research Survey</i> , <b>2024</b>                                | Reviews how AI and Machine Learning can improve Wi-Fi performance, resource allocation, and QoS                                     |



RV College of  
Engineering®

# Literature Survey

*Go, change the world®*

| Sl.No. | Paper Title, Journal Name,<br>Year                                                                                                                                        | Summary                                                                                                   |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 5.     | <b>The Path Towards Virtualized Wireless Communications: A Survey and Research Challenges</b> , <i>Journal of Network and Systems Management (Springer)</i> , <b>2023</b> | Reviews virtualization, resource allocation, and QoS management in future wireless communication systems. |



RV College of  
Engineering®

*Go, change the world®*

# Literature Gap

- Existing studies mainly focus on complex AI/Deep Learning models, which require high computational resources.
- Many works focus only on QoS prediction and do not provide adaptive bandwidth allocation recommendations.
- No studies compare Decision Tree and Random Forest for Wi-Fi QoS prediction.
- Existing systems often lack an interactive real-time dashboard for live network monitoring and prediction.
- There is a need for a simple, lightweight, and cost-effective ML-based solution for adaptive Wi-Fi bandwidth allocation.



RV College of  
Engineering®

*Go, change the world®*

# Novelty

- Adaptive Wi-Fi bandwidth allocation based on real-time network conditions rather than static QoS prediction
- Uses three key QoS parameters (Delay, Packet Loss, and Throughput) for efficient network priority classification.
- Real-time Streamlit dashboard for network monitoring, priority prediction, and bandwidth recommendation.



RV College of  
Engineering®

*Go, change the world®*

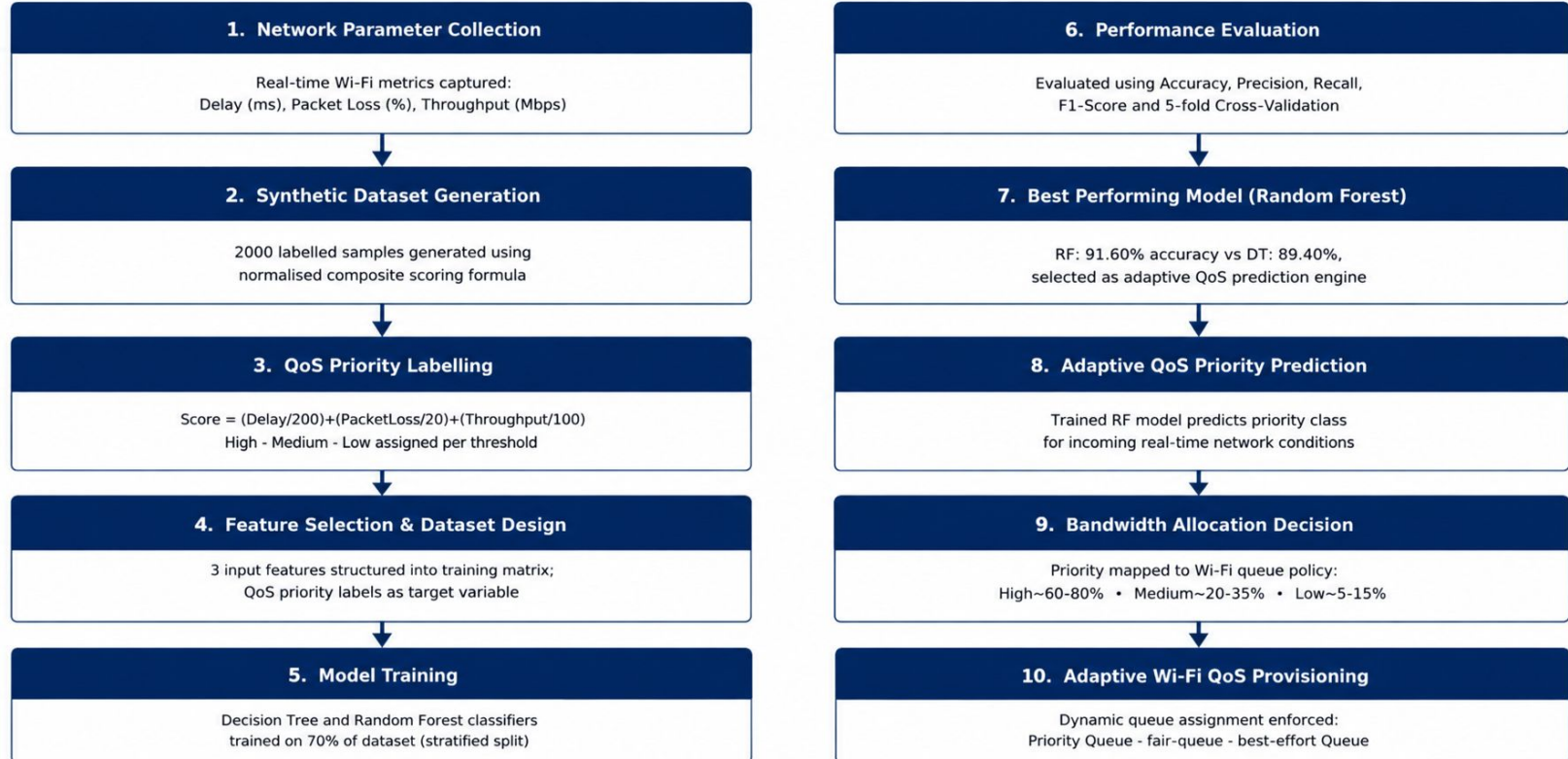
# Objectives

- Develop an adaptive Wi-Fi bandwidth allocation system using Machine Learning.
- Predict network priority (High, Medium, and Low) based on Delay, Packet Loss, and Throughput.
- Compare the performance of Decision Tree and Random Forest algorithms for QoS prediction.
- Design an interactive Streamlit dashboard for real-time prediction, visualization, and performance analysis.



# Methodology

*Go, change the world®*





RV College of  
Engineering®

# Software/Tool

*Go, change the world®*

| Software/Tool                | Purpose                                                         |
|------------------------------|-----------------------------------------------------------------|
| Python 3.x                   | Core programming language for implementation                    |
| Visual Studio Code (VS Code) | Code development and debugging                                  |
| Streamlit                    | Interactive web dashboard development                           |
| Scikit-learn                 | Machine learning model training (Decision Tree & Random Forest) |
| NumPy                        | Numerical computations and dataset generation                   |
| Pandas                       | Data manipulation and preprocessing                             |
| Matplotlib                   | Data visualization and graph plotting                           |
| Seaborn                      | Statistical data visualization                                  |



# Outcomes

*Go, change the world®*

## Front end





# Results and Discussions

*Go, change the world®*





RV College of  
Engineering®

# Metrics Table

*Go, change the world®*

| Metric                       | Decision Tree           | Random Forest       | $\Delta$ (RF – DT) |
|------------------------------|-------------------------|---------------------|--------------------|
|                              | — Overall —             |                     |                    |
| Test Accuracy                | 0.8940                  | 0.9460              | +0.0520            |
| CV Accuracy (mean $\pm$ std) | 0.8710 $\pm$ 0.0115     | 0.9290 $\pm$ 0.0174 | +0.0580            |
|                              | — Macro Averages —      |                     |                    |
| Macro Avg Precision          | 0.8848                  | 0.9399              | +0.0552            |
| Macro Avg Recall             | 0.8770                  | 0.9371              | +0.0601            |
| Macro Avg F1-Score           | 0.8804                  | 0.9385              | +0.0580            |
|                              | — Weighted Averages —   |                     |                    |
| Weighted Avg Precision       | 0.8945                  | 0.9458              | +0.0513            |
| Weighted Avg Recall          | 0.8940                  | 0.9460              | +0.0520            |
| Weighted Avg F1-Score        | 0.8938                  | 0.9459              | +0.0520            |
|                              | — Per-Class F1-Score —  |                     |                    |
| High Priority F1-Score       | 0.9064                  | 0.9489              | +0.0425            |
| Medium Priority F1-Score     | 0.7938                  | 0.8941              | +0.1003            |
| Low Priority F1-Score        | 0.9412                  | 0.9724              | +0.0312            |
|                              | — Per-Class Precision — |                     |                    |
| High Priority Precision      | 0.9380                  | 0.9559              | +0.0179            |
| Medium Priority Precision    | 0.7907                  | 0.8976              | +0.1069            |
| Low Priority Precision       | 0.9256                  | 0.9662              | +0.0406            |



# References

1. Alvarado, R., Ospina, B., Tellez, J., & Triana, V., "Dynamic Bandwidth Allocation with Machine Learning in Dense WiFi Network," *FACETS*, vol. 10, pp. 1–9, 2025.
2. Hao, X., She, C., Yeoh, P. L., Liu, Y., Vucetic, B., & Li, Y., "Hybrid-Task Meta-Learning: A GNN Approach for Scalable and Transferable Bandwidth Allocation," *IEEE Transactions on Wireless Communications*, vol. 23, no. 12, pp. 19820–19835, 2024.
3. Wilhelmi, F., Szott, S., Kosek-Szott, K., & Bellalta, B., "Machine Learning & Wi-Fi: Unveiling the Path Towards AI/ML-Native IEEE 802.11 Networks," 2024.
4. Yi, P., Cheng, W., Ju, Z., Wang, J., Pan, J., Ouyang, Y., & Zhang, W., "Fair AI-STA for Legacy Wi-Fi: Enhancing Sensing and Power Management with Deep Q-Learning," 2024.
5. Szott, S., Kosek-Szott, K., Gawłowicz, P., Gómez, J. T., Bellalta, B., Zubow, A., & Dressler, F., "Wi-Fi Meets ML: A Survey on Improving IEEE 802.11 Performance with Machine Learning," 2021.
6. Formis, G., Ericson, A., Forsström, S., Thar, K., Cena, G., & Scanzio, S., "Improving Wi-Fi Network Performance Prediction with Deep Learning Models," 2025.
7. Mawlood, M. A., & Mahmood, D. A., "Optimizing Weighted Fair Queuing with Deep Reinforcement Learning for Dynamic Bandwidth Allocation," *Telecom*, vol. 6, no. 3, 2025.
8. IEEE, IEEE 802.11™ Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications, IEEE Standard, latest revision



RV College of  
Engineering®

*Go, change the world®*

# THANK YOU